

BBO capstone project presentation

1. An overview of your BBO approach

- Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
- Briefly describe your overall strategy and how your process fits together.

Core objective

The Black Box Optimisation (BBO) capstone project tackles one of the most common challenges in applied machine learning: how to find the best next query points to an unknown, expensive-to-evaluate function using as few queries as possible. The function's internal mechanics are hidden — only inputs and their corresponding outputs are observable. The goal is to identify the input configuration that maximises the output across eight distinct functions (F1–F8), spanning two to eight input dimensions.

In a simple manner, the task is to learn to assess and analyse evidence provided by the model to apply the learnings to generate the next query point. Each query costs resource, so every new data point must be chosen to maximise the information it contributes about where the true optimum might lie.

Overall strategy

The approach is built around a Gaussian Process (GP) surrogate model — a probabilistic model that learns both the expected output and the uncertainty around that prediction across the entire input space. Rather than searching randomly or exhaustively, the GP is used to build an increasingly accurate map of the landscape, and a mathematical acquisition function (Upper Confidence Bound, UCB) guides where to query next by balancing two priorities:

- Exploration — querying in regions the model is uncertain about, in case the optimum lies there.
- Exploitation — querying near regions the model already believes are promising.

Layered on top of this GP-UCB foundation are three structural additions: (1) automatic kernel selection via cross-validation to ensure the GP fits the data well; (2) cluster-aware acquisition that identifies natural groupings in the observation space and directs search toward the most promising clusters; and (3) a suite of PCA-analogy enhancements that orient the search along the axes of maximum information — the directions where the model is most uncertain or where the data varies most meaningfully.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

Key changes in approach : From added complexity to principled simplicity

The most important strategic shift has been a deliberate move away from model complexity toward evidence-driven simplicity. Early iterations added components — Optuna hyperparameter optimisation for the GP, a CNN gradient predictor, and a residual corrector — based on the intuition that richer models would generalise better. Systematic evaluation across F1–F8 disproved this: HPO improved the cross-validated GP score by less than 0.001 on all eight functions, and the CNN stack consumed more than ten times the runtime of a plain GP without measurable gain in output quality. Based on this, Optuna HPO and CNN stack were removed in BBO11.

What replaced them was a more structured use of the information already available. The silhouette score gate (threshold 0.22) ensures that KMeans clustering is only activated when the data genuinely contains separable groups — not imposed on a featureless landscape. F8, with 50 observations in 8D, correctly has clustering bypassed entirely under this rule.

Version	Approach	Motivation	Result
BBO1-5	GP - Assess EI/UCB/TS approach / Support Vector like points	Initial assessment to identify high confidence regions	Stronger exploration / Exploration vs exploitation assessment
BBO6,7 and 10*	Gaussian Process surrogate with CNN augmentation and Optuna-driven HPO	More complexity assumed = better fit	<0.001 CV gain; CNN added 10x runtime cost
BBO8 and 9	LLM-based approach in parallel before returning to the GP framework in BBO10	Assess use of LLMs for BBO	LLM optimised approach, with new points targeting high-uncertainty, high-potential regions of each function's landscape
BBO11	Removed HPO and CNN; added silhouette gate for clustering	Data showed diminishing returns from both components	Faster pipeline; clustering conditional on signal quality
BBO12 (current)	Added E1–E5: ARD extraction, PCA pre-projection, uncertainty gradient, PCA candidates	PCA analogy: search along axes of maximum information	F5 confirmed 96% improvement; F8 clustering now viable

*BBO8 and BBO9 explored an LLM-based approach in parallel before returning to the GP framework in BBO10

Guiding principles now in place

- **Evidence-first.** Only trust components that evidence supports — HPO and CNN were removed because the data showed no gain.

- **Conditional activation.** Clustering activates only when silhouette ≥ 0.22 ; gradient steps only when they improve UCB.
- **Automatic relevance.** ARD kernels and PCA pre-projection reveal which dimensions matter without manual feature selection.
- **Efficient enhancements.** The uncertainty gradient (E1) and PCA-stratified candidates (E4) add intelligence without extra GP evaluations.

3. Patterns, data and insights

- Highlight the most meaningful trends you've observed across your data points or model evaluations.
- Identify which variables or behaviours seem to influence results the most.
- Explain how these observations have shaped your understanding of the search process.

Function-level diagnostic summary

Across 263 total observations, clear structural patterns have emerged. The table below summarises the key diagnostic signals for each of the eight functions.

Fn	Dim	n	Silhouette	ARD top	PC1-y	Landscape notes
F1	2D	21	0.53 — Strong	d1 (89%)	0.14	Compact; 95% near best
F2	2D	21	0.47 — Strong	d0 (100%)	0.28	Effectively 1-dimensional
F3	3D	26	0.36 — Moderate	d2 (91%)	0.60	Single axis drives output
F4	4D	41	0.27 — Moderate	d2 (25%)	0.75	Balanced; all dims matter
F5	4D	31	0.30 — Moderate	d3 (34%)	0.58	Sharp peak — 96% gain found
F6	5D	31	0.21 — Skipped	d1 (27%)	0.75	Diffuse; pure UCB used
F7	6D	41	0.23 — Weak	d4 (53%)	0.32	Sparse; d4 is key signal
F8	8D	51	0.13 — Skipped	d2 (20%)	0.93	Isotropic — PCA pre-projection applied

Most influential findings

F5 — This contains the most sharply localised peak across all eight functions. The BBO11 suggested query point returned an output of 8,662. F5 has only 3% of observations within 10% of the optimum, confirming a sharply localised peak that uniform sampling was unlikely to find.

F8 — PC1 of F8's input space correlates at -0.93 with the output — a single dominant direction. Yet KMeans silhouette is only 0.13 because the raw input variance is spread nearly uniformly across all 8 dimensions. The predictive structure only becomes visible after PCA pre-projection.

F2 — The ARD kernel assigns a length scale of 20 (the upper bound) to dimension d1, functionally removing it. F2 is a 2D function that behaves as 1D. This is directly analogous to PCA discarding a near-zero eigenvalue — the GP learns it automatically.

Across the full set, functions divide into three regimes that warrant different search strategies. Low-dimensional with strong cluster structure (F1, F2, F3) favour tight exploitation. Moderate-dimensional with partial structure (F4, F5, F7) require a balanced exploration-exploitation blend. High-dimensional near-isotropic functions (F6, F8) rely primarily on GP uncertainty rather than any geometric structure.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

Balancing exploration and exploitation

The UCB acquisition function controls the exploration-exploitation balance through its kappa parameter, fixed at 4.0. Empirical analysis showed that optimising kappa produced less than 0.003 improvement in leave-one-out rank score across F1–F8 — not worth the overhead of another optimisation loop. A fixed kappa = 4.0 provides sufficient exploration weight to avoid premature convergence while still directing queries toward high-mean regions.

Where clustering is active, the acquisition blend weight (cluster_acq_weight = 0.25) means 75% of the signal comes from GP uncertainty and 25% from cluster promise scores. This ensures the cluster signal informs but does not dominate — a deliberate safeguard after early versions over-weighted cluster promise and generated queries in already-dense regions.

Two examples of strategic decisions — what worked and what did not

Removing the CNN components — what did not work

Earlier versions included a CNN to help predict and correct the GP's output. The assumption was that a more complex model would produce better results. In practice it did not — the CNN struggled with functions whose outputs vary at very different scales (F4 ranges across 56 units; F5 across 8,662), and the corrections it made were inconsistent. It also added ten times the runtime of the plain GP with no measurable improvement in result quality. Removing it made the pipeline faster and the acquisition recommendations more reliable.

The F5 query — what worked

BBO11's cluster-aware acquisition identified a gap between two groups of observations in F5's input space — a region that had not yet been sampled. Placing the next query in that gap, between two moderately promising clusters, returned an output of 8,662 against a previous maximum output of 4,414 — nearly doubling the known best result. The lesson is straightforward: when the best outcome is concentrated in a small region of the input space, the most valuable query is often in the unexplored area between promising groups, not near the known best point.

Handling uncertainty and unexpected results

When results fall below expectations — as with F6 and F8 where clustering was correctly bypassed — the response is not to force structure onto the data but to fall back to pure UCB with PCA-stratified candidates. The silhouette threshold acts as an automatic reset: rather than tuning clustering parameters when they produce weak groupings, the system simply disengages that component. This mirrors the PCA principle of discarding low-variance components rather than rescaling them to appear informative.

The uncertainty gradient step (E1) provides a final layer of adaptivity: when the optimiser settles on a locally promising point, a final step redirects the query toward the area where the model still has the most to learn — the direction most likely to reveal new information rather than confirm what is already known.

5. Next steps and reflection

- Outline your planned actions or experiments to improve the model's performance.
- Include two or three sentences connecting your work to the broader ML landscape.
- Add a short statement on how you'd communicate your results to a non- technical audience or stakeholder.

Next actions

The strategy for the next query round should be differentiated by function regime rather than applied uniformly across all eight functions:

- F5 has a confirmed sharp peak. The final query should exploit tightly around the BBO12-identified neighbourhood using low kappa and high cluster weight to maximise the chance of landing at or above the current best.
- F1, F2, F3 and F6 had their initial designs already locate the best known points. The uncertainty gradient step should steer the final query toward the remaining unexplored corners rather than near the known best, prioritising coverage over refinement.
- F7 and F8 remain the highest-uncertainty functions. Both warrant continued broad UCB exploration with E3 PCA pre-projection active to compress the effective search space before any clustering decisions are made.
- F4 has uniform ARD weights, suggesting a genuinely multi-modal landscape without a dominant direction. The final query here carries the most strategic risk; retaining the full kappa = 4.0 exploration bonus is the appropriate conservative choice.

Connection to the broader ML landscape

Bayesian optimisation is increasingly central to applied machine learning, most visibly in neural architecture search and large-scale hyperparameter tuning where each evaluation is expensive. The GP-UCB framework used here is a direct precursor to the acquisition strategies now embedded in tools like Google Vizier. The shift from HPO-heavy to evidence-driven, lightweight pipelines mirrors a broader trend in the field: the most robust models tend to be those that correctly identify what the data justifies, not those with the most parameters. The PCA-analogy enhancements developed in BBO12 — particularly PCA pre-projection for high-dimensional spaces and ARD weight extraction — reflect ideas that underpin active learning and experimental design more broadly.

Communicating to a non-technical audience

Black box optimisation is a way of finding the best solution when the inner workings of a system are unknown. It improves results by learning from each attempt—balancing refinement of what works (exploitation) with trying new options (exploration), like tuning a machine through changing settings, observing the results, and use what you learn to improve the next attempt. The goal is to find the best result in as few tries as possible.

This project applied that approach across eight different “machines” (F1–F8) of varying complexity. In most cases, the method quickly learned which settings mattered most and focused the search on the most promising areas.

The strongest result was in Function 5, where the method found an output value that was almost twice as the maximum value obtained from previous outputs, showing it can discover solutions that random search misses.

Overall, the approach is faster and more efficient than random trial-and-error, because it learns as it goes and focuses effort where it is most likely to pay off.

